

Math 147

Analysis

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1 Course Notes

1.1 General Information

1.1.1 Contact

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Office hours: Monday, 3:30–4:30 p.m., and Friday, 10:00–11:00 a.m., in MC 4006.

1.1.2 Course Structure

Lectures: Section 002 meets on Monday, Wednesday, and Friday from 12:30–1:20 p.m. in MC 4021.

2 Introduction to Proof

2.1 Lecture 1

2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

2.1.2 Numbers

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1 (Irrationality of $\sqrt{2}$). *The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.*

Proof. Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 2n^2.$$

Thus, m^2 is even, so m is even. Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned}2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2.\end{aligned}$$

Therefore, n^2 is even, and hence n is even as well.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.2 (Decimal representation). Every real number has a decimal representation. In general, a decimal expansion can be written as

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{Z}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Example 2.3 (Recurring decimal). Is the statement $1.999\cdots = 2$ correct?

Yes. Let $x = 1.999\cdots$. Then

$$\begin{aligned} x &= 1.999\cdots, \\ 10x &= 19.999\cdots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

2.1.3 Statements and Quantifiers

Definition 2.4 (Statement). A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 2.5. The statement

$$\forall x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 2.6. The statement

$$\exists x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is true; for instance, $x = 2$ satisfies the inequality.

Proposition 2.7. *For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that*

$$x^3 - y^3 = 1.$$

Proof. Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Proposition 2.8. *The statement*

$$\exists y \in \mathbb{R}, \quad \forall x \in \mathbb{R}, \quad x^3 - y^3 = 1$$

is false.

Its negation is

$$\forall y \in \mathbb{R}, \quad \exists x \in \mathbb{R}, \quad x^3 - y^3 \neq 1.$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark. The order of quantifiers matters.

2.2 Mathematical Induction

Definition 2.9 (Principle of Mathematical Induction). Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true;

then

$$\forall n \in \mathbb{N}, \quad P(n)$$

is true.

Proposition 2.10 (Sum of the First n Natural Numbers). *For every $n \in \mathbb{N}$,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1+1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, $P(k+1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 2.11. *Let $S \subseteq \mathbb{N}$. If*

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k + 1 \in S$;

then $S = \mathbb{N}$.

Proof. Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k+1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

2.2.1 Well-Ordering Principle

Theorem 2.12 (Well-Ordering Principle). *Every nonempty subset S of $\mathbb{N}$ has a smallest element.*

Proof. Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

Since S has no smallest element, $1 \notin S$, so $1 \in T$. Suppose $k \in T$. If $k+1 \notin T$, then $k+1 \in S$, while no element of $\{1, 2, \dots, k\}$ belongs to S . Thus, $k+1$ would be the smallest element of S , a contradiction. Therefore, $k+1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so $S = \emptyset$, contradicting the assumption that S is nonempty. Therefore, S has a smallest element. ■